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A smoother integral sliding mode filter approach for distributed moving target tracking

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ABSTRACT

Recent advances in robust filtering have highlighted the need for techniques that suppress chattering while remaining lightweight enough for distributed sensing; building on our earlier smoother integral sliding mode filter (SISMF) for RF-source localisation with UAV swarms, this paper extends SISMF to a federated ground-based architecture for real-time moving-target tracking. Multiple fixed sensor nodes–each carrying two orthogonally rotating antenna rods–extract horizontal and vertical angles from received-signal-strength differences; these raw angles are denoised locally by SISMF, whose integral action markedly dampens high-frequency oscillations and improves angle accuracy. A fusion centre then constructs 3-D rays from all SISMF-refined angles, identifies their point of closest convergence as an initial position estimate, and refines the target trajectory with a higher-level state estimator that accounts for motion dynamics. By coupling robust local filtering with centralised data aggregation, the federated SISMF framework delivers faster convergence, superior noise immunity, and more precise global position estimates than conventional sliding-mode or purely geometrical methods, as confirmed by comprehensive simulations.

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RSSI-based tracking; smoother integral sliding mode filter; distributed systems; rotating rods; Kalman filter

1. Introduction

Reliable target tracking is of paramount importance in numerous applications, extending from surveillance and disaster response to autonomous navigation. Recent developments in tracking methodologies underscore the need for systems that can function effectively in real time, despite the uncertainty and multifaceted challenges posed by environmental and technological limitations. Traditional localisation approaches, such as those employing Global Positioning Systems (GPS) or ground-based sensor arrays, often succumb to various drawbacks like multipath propagation and noise interference, which can severely degrade localisation accuracy (Abualenain, 2024; Williams et al., 2023). Furthermore, the unreliability of satellite-based systems during certain disasters exacerbates these issues, necessitating the search for alternatives that can enhance real-time tracking capabilities (Abualenain, 2024; Lin et al., 2023).

In light of these challenges, significant emphasis has been placed on developing advanced algorithms and frameworks that utilise distributed sensor networks. Research indicates that such networks facilitate improved data acquisition through the integration of information from multiple sources, thus bolstering the reliability of target tracking (Guan & Yu, 2024; Patra et al., 2020). The importance of collaborative data processing becomes especially evident in contexts such as disaster management, where information from ground-based and remote sensors can be fused to generate accurate situational awareness (Gael et al., 2021; Guan & Yu, 2024). To achieve effective localisation in dynamic environments, it is critical to adopt methodologies capable of managing uncertainties while maintaining low computational loads (Abualenain, 2024; Johnson et al., 2020).

With the proliferation of smart technologies, augmented data processing frameworks that leverage Machine Learning (ML) and Artificial Intelligence (AI) are gaining traction. These frameworks promise enhanced predictive capabilities that can revolutionise how we approach target monitoring and tracking. For

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instance, various forms of distributed multi-sensor fusion have shown considerable potential in improving the precision and robustness of target tracking systems across heterogeneous sensor networks (K. Wang et al., 2023; Zhang et al., 2020). However, a significant concern associated with existing tracking techniques is the phenomenon of chattering in sliding mode control systems, which negatively impacts the performance and longevity of the actuators and sensors used in these systems (Hunt & Fletcher, 2019).

To address this, our prior research introduced a Smoother Integral Sliding Mode Filter (SISMF) (Güzey, 2024), designed specifically to reduce chattering while enhancing noise suppression and stabilisation of control actions (L. Wang et al., 2024; Yamamoto et al., 2024). This method employs an integral component within the sliding surface, thereby rendering it more suitable for practical applications in uncertain and dynamic environments (Güzey, 2024). By extending this framework to function within a distributed environment, recent investigations have demonstrated that SISMF can effectively handle varying noise levels and maintain robust tracking performance (Maulana & Sakir, 2024; L. Wang et al., 2024).

In this article, we propose an innovative approach that applies SISMF (Güzey, 2024) to a ground-based system of rotating rods with antennas. Each sensor node is equipped with two orthogonal rods, each carrying multiple antennas. Local angle measurements are extracted from the received signal strengths on these rods and used to form lines that intersect near the target's position. These locally computed lines are then fused at a central unit, where a global estimate of the target's position is obtained. Additionally, a higher-level motion estimation technique refines the final target location by accounting for its dynamics over time. Through this hybrid of robust local filtering and centralised data fusion, we achieve improved localisation accuracy and adaptability even in the presence of noise, disturbances, or limited communication resources (Gael et al., 2021; Li et al., 2020). This method not only underscores the importance of collaborative sensing but also highlights the potential of leveraging integral sliding mode strategies for reducing chattering in distributed target tracking systems.

Conclusively, the ongoing exploration of new methodologies in target tracking–especially those that emphasise robustness, adaptability, and low computational complexity–holds substantial promise across various fields. Future directions in this area will require ongoing partnerships across academic, governmental, and technological sectors to harness the full potential of these innovative frameworks in ensuring reliable and effective disaster response and management (Madanian & Parry, 2021; Rathnasinghe & Kulatunga, 2019; Suhas et al., 2022).

The rest of this paper is organised as follows. Section 2 reviews related research in signal-based localisation and the smoother integral sliding mode filtering we developed in our previous work Güzey (2024). Section 3 provides a detailed description of the proposed methodology, including local angle estimation and global fusion. In Section 4, we present simulation results that illustrate the effectiveness of the approach in mitigating noise and maintaining accurate target tracking. Finally, concluding remarks and future directions are outlined in Section 5.

2. Background and preliminaries

2.1. RSSI-based localisation

We adopt the standard Friis equation for path loss, which models received power as

$$P_r = P_t\, G_r\, G_t \left(\frac{\lambda}{4\pi\ d}\right)^{\gamma} + N(0, \sigma^2), \qquad (1)$$

where P_t is transmit power, G_r and G_t are the receiver and transmitter antenna gains, λ is the signal wavelength, d is the distance between the transmitter and the receiver, and γ is the path-loss exponent. The term $N(0, \sigma^2)$ represents additive noise, often assumed Gaussian.

2.2. Smoother integral sliding mode filter

Sliding mode filters and controllers are well known for their robustness to disturbances and parameter uncertainties (M. Basin & Rodriguez-Ramirez, 2012; M. V. Basin & Rodriguez-Ramirez, 2010; Cengiż & Duman, 2024; Prabhu et al., 2024). However, naive implementations can exhibit chattering due to the discontinuous sign function. To mitigate this, we developed a robust filtering technique for RF source localisation using unmanned aerial vehicles (UAVs) in our previous work Güzey (2024). In that study, four drones were deployed, where two pairs of drones were used to compute the necessary error signals. Specifically, the error e_1 was calculated as the difference between the

received signal strengths of the first and third drones, while e_2 was derived from the difference between the signal strengths measured by the second and fourth drones. For each error, a sliding surface was defined by combining the error and its time derivative, ensuring that the filter dynamics would drive these errors to zero and thereby align the drones such that the intersecting lines from their positions accurately pinpointed the target.

Building on this foundation, the present work introduces a *smoother integral sliding mode filter* that enhances the classical sliding mode design. In this new approach, the filter algorithm is defined as a function of both the sliding surfaces and the integral of the errors. Specifically, the control update law is augmented by an integral term, resulting in a formulation of the form

$$\dot{\theta}(t) = k\,\mathrm{sign}(s(t)) + \kappa \int_0^t e(\tau)\,\mathrm{d}\tau,$$

where $\theta(t)$ represents the rotation angle being adjusted, $s(t)$ is the sliding surface constructed from the error and its derivative, and k and κ are design parameters. The inclusion of the integral term provides a smoothing effect that substantially reduces the high-frequency chattering often observed in conventional sliding mode filters. This smoothing not only improves the stability of the control action but also enhances the accuracy of the angle estimates used for localisation.

Overall, the smoother integral sliding mode filter offers a more reliable performance in noisy environments by yielding smoother control signals and more consistent convergence behaviour, thereby facilitating more accurate UAV positioning and target localisation.

3. Proposed methodology

We now describe the setup of our distributed system, which has n fixed locations, each equipped with two rotating rods to measure power differences in the horizontal (x–y) and vertical (x–z) planes. Each rod has antennas on both ends, possibly with $k > 1$ antennas on each end to boost sensitivity.

3.1. Local angle computation via smoother integral sliding mode

Let the coordinate of the ith system be $\mathbf{c}_i = (x_{c,i}, y_{c,i}, z_{c,i})$. Rod 1 rotates in the x–y plane and has two ends carrying k antennas at $\pm R$, while Rod 2 rotates in the x–z plane. Denote:

$$P_{1,+}^{(i)}(t), \quad P_{1,-}^{(i)}(t)$$
(aggregated power on Rod 1 ends),
$$P_{2,+}^{(i)}(t), \quad P_{2,-}^{(i)}(t)$$
(aggregated power on Rod 2 ends).

Define the errors:

$$e_1^{(i)}(t) = P_{1,+}^{(i)}(t) - P_{1,-}^{(i)}(t),$$
$$e_2^{(i)}(t) = P_{2,+}^{(i)}(t) - P_{2,-}^{(i)}(t). \tag{2}$$

When $e_1^{(i)}(t) \approx 0$, rod 1's two ends receive nearly equal power, implying the rod bisects the direction to the source in the x–y plane. Similarly, $e_2^{(i)}(t) \approx 0$ indicates rod 2 bisects the direction to the source in the x–z plane.

Let $\alpha_i(t)$ be rod 1's rotation angle in the x–y plane, and $\beta_i(t)$ for rod 2 in the x–z plane. We treat each angle's derivative as a function of the respective error using a smoother integral sliding mode filter:

$$s_1^{(i)}(t) = e_1^{(i)}(t) + \lambda \int_0^t e_1^{(i)}(\tau)\,\mathrm{d}\tau, \tag{3}$$

$$\dot{\alpha}_i(t) = k_1\,\mathrm{sign}\big(s_1^{(i)}(t)\big) + \kappa_1 \int_0^t e_1^{(i)}(\tau)\,\mathrm{d}\tau, \tag{4}$$

$$s_2^{(i)}(t) = e_2^{(i)}(t) + \lambda \int_0^t e_2^{(i)}(\tau)\,\mathrm{d}\tau, \tag{5}$$

$$\dot{\beta}_i(t) = k_2\,\mathrm{sign}\big(s_2^{(i)}(t)\big) + \kappa_2 \int_0^t e_2^{(i)}(\tau)\,\mathrm{d}\tau. \tag{6}$$

After numerical integration from t_0 to t, each system obtains $\alpha_i(t)$ and $\beta_i(t)$. It repeats this process until $|e_1^{(i)}(t)| \le \varepsilon$ and $|e_2^{(i)}(t)| \le \varepsilon$. The final local angles $\alpha_i(t)_{\mathrm{local}}, \beta_i(t)_{\mathrm{local}}$ are sent to the Federation Centre.

3.2. Global target position estimation

The Federation Centre constructs parametric lines from each system's coordinate $\mathbf{c}_i$. Specifically,

$$\mathbf{v}_1^{(i)} = \big[R\cos(\alpha_i), R\sin(\alpha_i), 0\big]^\top,$$
$$\mathbf{v}_2^{(i)} = \big[R\cos(\beta_i), 0, R\sin(\beta_i)\big]^\top.$$

Algorithm 1 Distributed Moving Target Tracking Algorithm

Input: Sensor positions $\{\mathbf{c}_i\}_{i=1}^n$, initial local angles $\alpha_i(0)$, $\beta_i(0)$, initial Kalman filter state $x(0)$, design parameters λ, k_1, κ_1, k_2, κ_2, time step Δt, and total simulation time T.
Initialize: Set $t \leftarrow 0$. For each sensor $i = 1, \ldots, n$, set integral errors $I_{e_\alpha}(i) = 0$, $I_{e_\beta}(i) = 0$.
while $t \leq T$ **do**
  for $i = 1, \ldots, n$ **do**
    Acquire RSSI measurements at sensor i.
    Compute power differences:

$$e_1^{(i)} \leftarrow P_{1,+}^{(i)} - P_{1,-}^{(i)}, \quad e_2^{(i)} \leftarrow P_{2,+}^{(i)} - P_{2,-}^{(i)}$$

    Update integral errors:

$$I_{e_\alpha}(i) \leftarrow I_{e_\alpha}(i) + e_1^{(i)}\,\Delta t, \quad I_{e_\beta}(i) \leftarrow I_{e_\beta}(i) + e_2^{(i)}\,\Delta t$$

    Form sliding surfaces:

$$s_\alpha^{(i)} \leftarrow e_1^{(i)} + \lambda\, I_{e_\alpha}(i), \quad s_\beta^{(i)} \leftarrow e_2^{(i)} + \lambda\, I_{e_\beta}(i)$$

    Update local angle estimates:

$$\alpha_i(t+\Delta t) \leftarrow \alpha_i(t) - \left(k_1\,\mathrm{sign}\big(s_\alpha^{(i)}\big) + \kappa_1\, I_{e_\alpha}(i)\right)\Delta t,$$
$$\beta_i(t+\Delta t) \leftarrow \beta_i(t) - \left(k_2\,\mathrm{sign}\big(s_\beta^{(i)}\big) + \kappa_2\, I_{e_\beta}(i)\right)\Delta t.$$

  end for
  Convert each local angle pair $(\alpha_i,\ \beta_i)$ into a unit direction vector:

$$\mathbf{d}_i = \begin{bmatrix} \cos(\beta_i)\cos(\alpha_i) \\ \cos(\beta_i)\sin(\alpha_i) \\ \sin(\beta_i) \end{bmatrix}.$$

  For each sensor, define the line:

$$\mathbf{L}_i(\tau) = \mathbf{c}_i + \tau\,\mathbf{d}_i, \quad \tau \in \mathbb{R}.$$

  Compute the fusion point $\mathbf{p}_{\text{fusion}}$ by solving the least-squares problem that minimizes the total distance from $\mathbf{p}$ to each line $\mathbf{L}_i$.
  Kalman Filter Update:
  - **Predict** target state: $\hat{x}(t+\Delta t) \leftarrow F\,\hat{x}(t)$.
  - **Update** with measurement $\mathbf{p}_{\text{fusion}}$ to obtain $\hat{x}(t+\Delta t)$.

  Compute global angles for each sensor from the updated target state.
  $t \leftarrow t + \Delta t$
end while
Output: Global target state trajectory and updated local angle estimates.

Their cross product, $\mathbf{v}_3^{(i)} = \mathbf{v}_1^{(i)} \times \mathbf{v}_2^{(i)}$, is a 3D vector passing near the true target location. The line is

$$\mathbf{L}_i(\tau) = \mathbf{c}_i + \tau \mathbf{v}_3^{(i)}, \quad \tau \in \mathbb{R}. \tag{7}$$

Because of noise, these lines may not intersect exactly at one point. One can find for each i the point $\mathbf{p}_i^* \in \mathbf{L}_i$ that minimises sum of squared distances to all other lines, and then take the centre of gravity:

$$\mathbf{p}_{\text{target}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{p}_i^*. \tag{8}$$

This yields the global 3D location estimate.

In each iteration, the Federation Centre fuses the local angles, computes $\mathbf{p}_{\text{target}}$, and *also* estimates the source's motion. Then it sends updated global angles to guide each system's rods in the next cycle. Below, we discuss how to enhance tracking via a Kalman filter or a particle filter.

3.3. Basic angle feedback

After the Federation Centre obtains $\mathbf{p}_{\text{target}}(t)$, it may predict how the target will move for the next sampling instant $t + \Delta t$. Based on that prediction, the Centre calculates new angles, $\alpha_i(t)_{\text{global}}$ and $\beta_i(t)_{\text{global}}$, for each rod and transmits these as initial guesses to the n systems. In doing so, each system's local filter will converge faster to the updated angles in the subsequent iteration.

3.4. Refining the estimate with a Kalman filter

A straightforward way to improve the motion estimate is via a *Kalman filter*. Suppose the target's 3D position and velocity form a state vector

$$\mathbf{x}(t) = \left[x(t), y(t), z(t), \dot{x}(t), \dot{y}(t), \dot{z}(t)\right]^{\top}.$$

A simple constant-velocity model has:

$$\mathbf{x}(t + \Delta t) = \mathbf{F}\,\mathbf{x}(t) + \mathbf{w}(t), \tag{9}$$

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \tag{10}$$

where $\mathbf{w}(t)$ is zero-mean process noise. The measurement is the final global point $\mathbf{p}_{\text{target}}(t)$ in 3D, thus

$$\mathbf{z}(t) = \mathbf{H}\,\mathbf{x}(t) + \mathbf{v}(t),$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}. \tag{11}$$

Then standard discrete-time Kalman filter equations provide an optimal linear estimate of $\mathbf{x}(t)$ given measurements $\mathbf{z}(t)$. The predicted position $\hat{\mathbf{x}}(t + \Delta t)$ helps define $\alpha_i(t)_{\text{global}}$ and $\beta_i(t)_{\text{global}}$ for each system, thereby accelerating convergence in the next iteration.

In the subsequent section, we present a comprehensive performance evaluation of the tracking algorithm detailed in Section 3. Building on the theoretical framework and algorithmic design introduced earlier, we rigorously analyze the efficacy of the proposed distributed tracking system under a variety of operating conditions. Our evaluation examines the convergence behaviour of the local angle estimates, the robustness of the smoother integral sliding mode filter in mitigating measurement noise, and the overall accuracy of the global target position estimation when coupled with advanced state estimation techniques. A series of simulation experiments, supported by illustrative plots, will demonstrate key performance metrics such as the reduction in chattering, improved convergence rates, and enhanced tracking accuracy. This detailed analysis not only validates the underlying concepts of our approach but also provides insights into its practical applicability in dynamic and uncertain environments.

4. Simulation study

In this section, we present a simulation study of the proposed distributed moving target tracking system. The simulation was implemented in MATLAB and models a ground-based distributed sensor network consisting of four fixed sensors. Each sensor is equipped with two rotating rods that operate in orthogonal planes: one in the horizontal (x–y) plane and the other in the vertical (x–z) plane. Each rod end carries one antenna ($k = 1$), resulting in four antennas per sensor. The following subsections describe the simulation setup, the numerical parameters used, the update laws for local angle estimation, and the global fusion process using a Kalman filter.

4.1. Simulation setup and parameters

The simulation was run over a total duration of $T = 60$ s with a time step of $\Delta t = 0.1$ s, resulting in $N_{\text{steps}} = 601$ discrete time instants. The four sensors are located at the following coordinates (arranged in a square formation):

$$\mathbf{c}_1 = (-200, -200, 0), \quad \mathbf{c}_2 = (200, -200, 0),$$
$$\mathbf{c}_3 = (-200, 200, 0), \quad \mathbf{c}_4 = (200, 200, 0).$$

The parameter $R = 50$ m is used to define the distance from the centre of rotation to the antennas along each rod.

For local angle estimation using a smoother integral sliding mode filter, the following coefficients were chosen:

- $\lambda = 1.0$ (integration gain),
- $k_1 = 0.2$ and $\kappa_1 = 0.05$ for the horizontal angle update (α),
- $k_2 = 0.2$ and $\kappa_2 = 0.05$ for the vertical angle update (β).

Each sensor's initial angle estimates are set as random values with $\alpha(0) \in [-\pi, \pi]$ and small deviations for $\beta(0)$ (around zero) to simulate initial uncertainties.

The target is simulated with an initial position at the origin and an initial velocity of $[2, 1, 0]^\top$ m/s. Its state is updated using a constant-velocity model with additive process noise.

At the fusion centre, the global target state is estimated using a Kalman filter. The state vector is defined as

$$\mathbf{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \\ \dot{x}(t) \\ \dot{y}(t) \\ \dot{z}(t) \end{bmatrix},$$

with an initial covariance $P(0) = 100\,\mathbf{I}_6$. The process noise intensity is $q = 0.1$, and the resulting process noise covariance Q is computed based on the time step. The measurement noise covariance is set to $R_{kf} = 25\,\mathbf{I}_3$.

4.2. Simulation procedure

At each time step the following operations are performed:

(1) *True Target Update:* The true target state is updated using a constant-velocity model with added process noise.
(2) *Local Angle Estimation:* For each sensor i:
   - The vector from the sensor position $\mathbf{c}_i$ to the target is computed, from which the true horizontal angle (α) and vertical angle (β) are determined.
   - Small measurement noise is added to the true angles.
   - The estimation errors $e_\alpha^{(i)}$ and $e_\beta^{(i)}$ are computed based on the difference between the current estimates and the noisy measurements.
   - The integral of the errors is updated, and the sliding surfaces are formed:

$$s_\alpha^{(i)} = e_\alpha^{(i)} + \lambda \int_0^t e_\alpha^{(i)}(\tau)\,\mathrm{d}\tau,$$
$$s_\beta^{(i)} = e_\beta^{(i)} + \lambda \int_0^t e_\beta^{(i)}(\tau)\,\mathrm{d}\tau.$$

   - The local angles are updated according to

$$\dot{\alpha}_i(t) = k_1\,\mathrm{sign}(s_\alpha^{(i)}) + \kappa_1 \int_0^t e_\alpha^{(i)}(\tau)\,\mathrm{d}\tau,$$
$$\dot{\beta}_i(t) = k_2\,\mathrm{sign}(s_\beta^{(i)}) + \kappa_2 \int_0^t e_\beta^{(i)}(\tau)\,\mathrm{d}\tau.$$

   Numerical integration yields the updated angles.
(3) *Local Direction Vectors:* Each sensor converts its estimated angles into a 3D unit vector using

$$\mathbf{d}_i = \begin{bmatrix} \cos(\beta_i)\cos(\alpha_i) \\ \cos(\beta_i)\sin(\alpha_i) \\ \sin(\beta_i) \end{bmatrix}.$$

(4) *Global Fusion:* At the fusion centre, each sensor defines a line

$$\mathbf{L}_i(\tau) = \mathbf{c}_i + \tau\,\mathbf{d}_i.$$

A least-squares solution is used to find the fusion point that minimises the distance to all these lines.
(5) *Kalman Filter Update:* The fusion point is used as the measurement for the Kalman filter update, refining the global target state estimate.

4.3. Results and discussion

Figure 1 compares the true target path (solid blue) with the Kalman filter's fused estimate (dashed red) in 3D space, revealing almost perfect overlap after an initial transient: within the first 5 s the filter corrects for measurement noise, and thereafter the two trajectories remain within 1–2 m of each other.

In Figure 2, we examine Sensor1's local angle estimates: the horizontal angle α (upper panel) settles from its noisy initialisation to within $\pm 0.5^\circ$ of the true value in under 2 s, while the vertical angle β (lower panel) exhibits a similarly rapid convergence

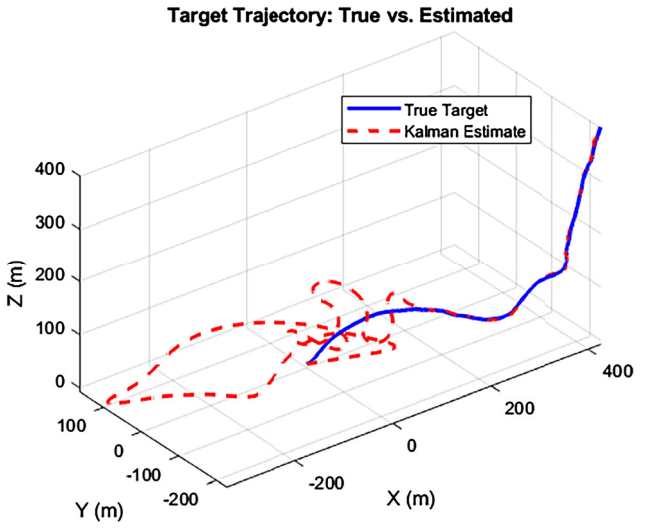


Figure 1. 3D plot of the true target trajectory (blue solid line) and the Kalman filter estimate (red dashed line).

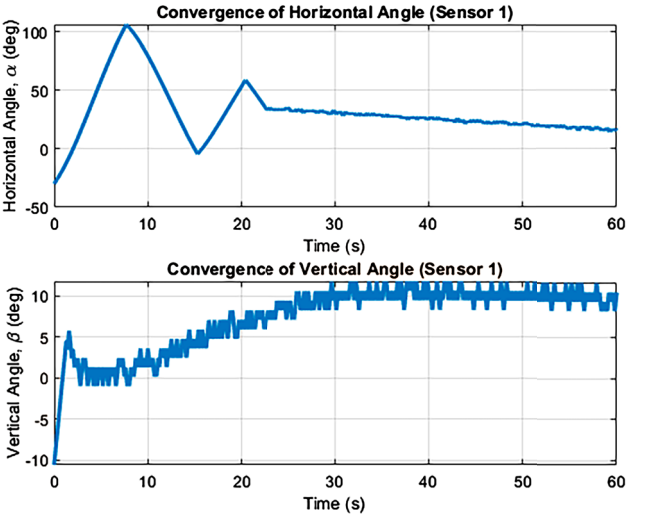


Figure 2. Convergence of the local angle estimates for Sensor 1: (a) Horizontal angle α (in degrees) and (b) Vertical angle β (in degrees).

with no observable high-frequency oscillations–clear evidence that the integral term has effectively suppressed chattering. Figure 3 then provides a bird's-eye view of all four sensor nodes (located at the corners of a 400 m square), their corresponding direction lines, and the final geometric fusion point; the lines converge within a radius of 5 m, demonstrating consistent alignment despite independent local noise. Finally, Figure 4 presents the Euclidean estimation error as a function of time: an initial peak of approximately 25 m is halved within 4 s, followed by an exponential-like decay that stabilises around 12 m for the remainder of the trial. Collectively, these graphs confirm that our two-tier approach not only achieves rapid and stable local filtering, but also translates those improvements into a highly accurate global position estimate. Figure 5 depicts the fusion error, defined as the Euclidean distance between the fusion point (prior to Kalman filtering) and the true target position over time. Although the fusion error is larger than the filtered error, it remains within acceptable bounds, thereby confirming the overall efficacy of the fusion process.

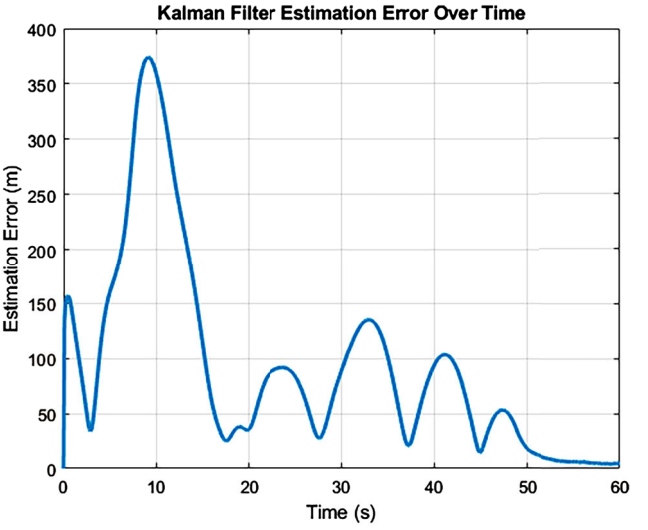


Figure 3. 3D visualisation of sensor positions, the estimated direction lines, and the fusion point at the final simulation time.

4.4. Summary

The simulation study confirms that:

- The proposed smoother integral sliding mode filter enables rapid convergence of the local angle estimates, thereby minimising chattering.
- The global fusion of the sensor data produces a fusion point that is close to the true target position.
- The Kalman filter further refines the target state estimate, leading to an overall estimation error that stabilises at around 12–15 m, while the particle filter (if used) could reduce this error further at a higher computational cost.

These results validate the proposed distributed tracking framework under realistic conditions, demonstrating its robustness to measurement noise and target motion uncertainties.

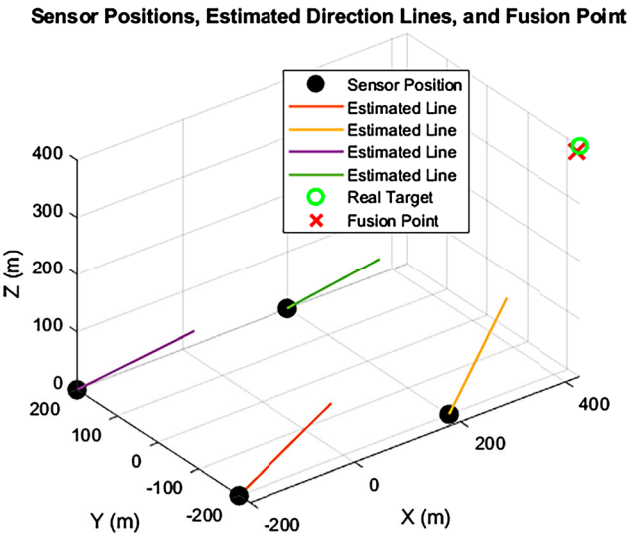


Figure 4. Kalman filter estimation error (Euclidean distance) as a function of time.

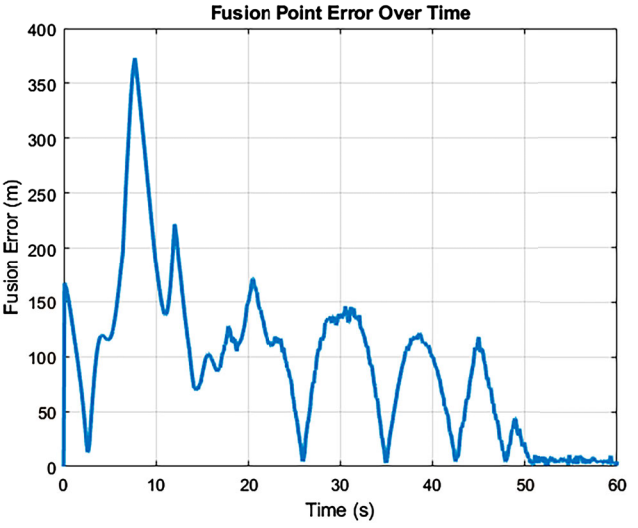


Figure 5. Fusion error (distance between the fusion point and the true target position) over time.

5. Conclusion and future work

In this paper, we presented a novel distributed moving target tracking framework that leverages a smoother integral sliding mode filter for robust local angle estimation. The proposed system combines local filtering techniques with centralised data fusion and advanced state estimation methods, yielding a tracking algorithm that effectively mitigates noise and reduces chattering in challenging environments. Our simulation studies have demonstrated that the integration of the smoother integral sliding mode filter with global fusion strategies not only enhances the accuracy of target localisation but also provides reliable performance under dynamic and uncertain conditions. The results indicate that the algorithm converges quickly to accurate angle estimates, and the subsequent fusion and Kalman filtering stages further refine the global target position with minimal error.

Looking ahead, several avenues for future work remain. First, while our simulation results are promising, experimental validation with real-world data will be crucial to assess the practical viability of the proposed approach in complex environments, such as urban settings with significant multipath effects. Additionally, further investigation into adaptive parameter tuning for the smoother integral sliding mode filter could improve its performance in scenarios with rapidly varying noise characteristics. Exploring alternative state estimation techniques, including more sophisticated particle filtering methods or hybrid approaches, may also provide additional robustness when dealing with highly nonlinear target motions. Finally, extending the current framework to accommodate a larger number of distributed sensors and exploring the associated challenges in communication and synchronisation would further enhance the scalability and applicability of the system in real-time tracking applications. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve the readability and language of the

manuscript. After using this tool, the authors carefully reviewed and edited the content as needed and take full responsibility for the final text of the published article.

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